

Notation

Typesetter's note: the source conventions below are presented with the modernizations specified in `NOTATION.md`. In particular, spacetime components are ordered $(0, 1, 2, 3)$ while retaining Weinberg's mostly-plus metric, states use Dirac notation, and Batalin–Vilkovisky antifields use $\ddagger$.

Latin indices $i, j, k, \dots$ generally run over the three spatial coordinate labels $1, 2, 3$. Where specifically indicated, they may run over the values $1, 2, 3, 4$, with the source convention $x^4 = it$.

Greek indices $\mu, \nu, \dots$ from the middle of the Greek alphabet generally run over the four spacetime coordinate labels $0, 1, 2, 3$, with x^0 the time coordinate.

Greek indices $\alpha, \beta, \dots$ from the beginning of the Greek alphabet generally run over the generators of a symmetry algebra.

Repeated indices are generally summed, unless otherwise indicated.

The spacetime metric is mostly plus,

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1).$$

The d'Alembertian is defined as

$$\square = \eta^{\mu\nu} \frac{\partial^2}{\partial x^\mu \partial x^\nu} = \nabla^2 - \frac{\partial^2}{\partial t^2},$$

where ∇^2 is the spatial Laplacian $\partial^2/\partial x^i \partial x^i$.

The Levi–Civita tensor $\epsilon^{\mu\nu\rho\sigma}$ is defined as the totally antisymmetric quantity with $\epsilon^{0123} = +1$.

Spatial three-vectors are indicated by letters in boldface. Three-vectors in isospin space are indicated by arrows.

A hat over any vector indicates the corresponding unit vector; thus $\hat{\mathbf{V}} = \mathbf{V}/|\mathbf{V}|$.

A dot over any quantity denotes the time derivative of that quantity.

Dirac matrices γ^μ are defined so that

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu}.$$

Also,

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3, \quad \beta = i\gamma^0.$$

The step function $\theta(s)$ has the value $+1$ for $s > 0$ and 0 for $s < 0$.

The complex conjugate, transpose, and Hermitian adjoint of a matrix or vector A are denoted A^* , $A^\top$, and $A^\dagger$, respectively. The Hermitian adjoint of an operator O is denoted $O^\dagger$, except where an asterisk is used to emphasize that a vector or matrix of operators is not transposed. $+H.c.$ or $+c.c.$ at the end of an expression indicates the addition of the Hermitian adjoint or complex conjugate of the foregoing terms. A bar on a Dirac spinor u is

defined by $\bar{u} = u^\dagger \beta$. The antifield of a field x^n in the Batalin–Vilkovisky formalism is denoted $x_n^\ddagger$, rather than x_n^* , to distinguish it from the ordinary complex conjugate or the antiparticle field.

Units are usually used with $\hbar$ and the speed of light taken to be unity. Throughout, $-e$ is the rationalized charge of the electron, so that the fine-structure constant is

$$\alpha = \frac{e^2}{4\pi} \simeq \frac{1}{137}.$$

Numbers in parentheses at the end of quoted numerical data give the uncertainty in the last digits of the quoted figure. Where not otherwise indicated, experimental data are taken from ‘Review of Particle Properties,’ *Phys. Rev.* **D50**, 1173 (1994).